

Context Compression Techniques for Legal Document Analysis

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Abstract

Legal application are bound to the handling of large scale documnt analysis when starting a case. The frontier models that powers the majority of LLM applications are reporting on increasing maximum sequece length / context windows. But the

Chapter 1

Introduction

The legal industry is increasingly adopting large language models (LLMs) to assist lawyers in navigating the complexity of modern litigation. Legal cases are inherently dynamic: they evolve over time as new documents, correspondence, evidence, and procedural developments continuously emerge. What was once a central claim may become peripheral; a newly submitted document may reframe the entire dispute. This progression means that not only does the volume of material grow, but the *relevance* and *weight* of existing information shifts as the case develops.

A fundamental limitation of current LLM-based systems is that they are stateless. Without explicit mechanisms for maintaining case state, models must re-read the full document history for every query. More critically, even models that handle large contexts tend to perform well in early stages but *drift* over extended, multi-turn interactions — gradually losing track of what is current, what has been superseded, and what actually matters. In legal case management, this drift is not merely an inconvenience; it can lead to factually incorrect summaries, missed deadlines, or misrepresentation of parties and claims.

This thesis addresses this problem through two complementary mechanisms: *compression* of case-relevant information into a structured representation, and *state maintenance* — the continuous, structured tracking of case essentials as the project evolves. The vehicle for this is a structured *FactSheet*: a living, machine-readable summary of the case that is initialized from initial input and updated incrementally as new documents and information arrive.

The research investigates whether maintaining an explicit, structured state representation improves an LLM agent’s ability to preserve contextual accuracy over the full lifecycle of a legal case — particularly as conversation length grows and the document landscape changes. The system is implemented as a LangGraph-based conversational agent, incorporating multi-phase document processing pipelines for both project initialization and incremental updates.

The contributions of this thesis are: (1) the design and implementation of initialization and update pipelines for structured legal case state management, (2) an evaluation framework comparing structured state agents against RAG-only and baseline approaches, and (3) an analysis of accuracy degradation, compression efficiency, and state consistency across extended multi-turn conversations.

The remaining chapters are organized as follows: Chapter 2 presents background and related work on long-context LLMs, context compression, and retrieval-augmented generation. Chapter 3 describes the methodology, system architecture, and evaluation design. Chapter 4 presents experimental results. Chapter 5 discusses findings and implications, and Chapter 6 concludes with a summary and directions for future work.

Chapter 2

Theory

This chapter presents the theoretical foundation for the approaches explored in this thesis. We examine the core challenges of applying large language models to long-context, multi-turn legal document analysis, and survey the relevant literature on retrieval, compression, and agentic context management strategies.

2.1 Large Language Models and Context Windows

Modern large language models (LLMs) are built on the transformer architecture [VSP⁺23], which processes input as a sequence of tokens bounded by a fixed context window. While recent models advertise longer context windows, empirical benchmarks reveal a significant gap between nominal and effective context utilization. introduce RULER, a benchmark designed to measure the *real* usable context size of long-context LLMs, demonstrating that performance degrades substantially before the advertised limit is reached [HSK⁺24]. This has direct implications for legal use cases, where case files can span hundreds of pages across dozens of documents.

2.2 The Lost in the Middle Problem

Even within the effective context window, LLMs do not attend uniformly to all input. Liu et al. show that model performance is highest when relevant information appears at the beginning or end of the context, and degrades for information placed in the middle — a phenomenon they term *lost in the middle* [LLH⁺23]. This positional bias has critical consequences for legal document analysis, where key facts may be embedded deep within lengthy attachments. It motivates the need for strategies that either restructure context placement or reduce reliance on verbatim document injection altogether.

2.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) addresses context limitations by retrieving only the most relevant document chunks at query time, rather than loading entire documents into the context window [LPP⁺21]. A standard RAG pipeline consists of offline chunking and embedding of documents into a vector store, followed by semantic retrieval at inference time. While effective for single-turn question answering, naive RAG struggles in multi-turn conversational settings where relevance depends on accumulated context across turns — a central challenge in this thesis.

RECOMP [XSC23] extends the RAG paradigm with both extractive and abstractive compressors that filter and condense retrieved passages before they are passed to the generator, reducing noise and token consumption while preserving relevant information.

2.4 Prompt Compression

An orthogonal approach to managing long contexts is prompt compression: reducing the token footprint of the input itself rather than selectively retrieving. LLMLingua [JWL⁺23] proposes a coarse-to-

fine compression method guided by a smaller language model, identifying and removing tokens with low information content. LongLLMLingua [JWL⁺24] extends this to long-document scenarios, incorporating question-aware compression that prioritizes tokens contextually relevant to the user query. Both methods are relevant as baseline comparisons for the context management strategies implemented in this thesis.

2.5 Graph-Based and Multi-Hop Retrieval

Standard vector similarity retrieval operates on isolated chunks and does not capture relational structure between entities. In legal cases, facts, parties, events, and claims are deeply interconnected, making relational retrieval particularly valuable. GraphRAG [ETC⁺25] proposes building a knowledge graph over the document corpus, enabling community-level summarization and structured traversal of entity relationships. HopRAG [LWC⁺25] further extends this with multi-hop reasoning, constructing logical chains across documents to answer complex queries that require synthesizing information from multiple sources. These approaches are particularly relevant to the factsheet extraction pipeline implemented in this thesis, where structured relationships between parties, events, and claims must be maintained across turns.

2.6 Infinite and Extended Context Architectures

An alternative to prompt-level strategies is modifying the underlying model architecture to support longer effective memory. Munkhdalai et al. propose Infini-Attention, which introduces compressive memory into the transformer attention mechanism to enable theoretically unbounded context without quadratic scaling [MFG24]. While this represents a promising long-term direction, it requires architectural changes to the model and is not applicable in API-based deployment settings, which is the constraint under which this thesis operates.

2.7 Agentic Context Management

Beyond static retrieval and compression, agentic systems can actively manage context across multi-turn conversations. Rolling summarization — periodically condensing conversation history into a compact summary — is a practical strategy for preventing context overflow in long sessions [Mic24]. Stateful agent frameworks such as LangGraph enable persistent state management through checkpointing, allowing the agent to maintain structured representations (e.g., factsheets) that accumulate case knowledge across turns without requiring the full document history to remain in context.

Taken together, the strategies surveyed in this chapter — effective context benchmarking, positional bias awareness, RAG with compression, graph-based retrieval, and agentic summarization — form the theoretical basis for the system design and experimental comparisons presented in the following chapters.

Chapter 3

Method

3.1 System Architecture

The system is designed as a modular, client-server application consisting of four main packages organized as a `uv` workspace: `agent`, `ui`, `evals`, and `data-viewer`. The architecture follows a separation of concerns, where the backend handles all agent logic, data persistence, and document processing, while the frontend is responsible for user interaction and result presentation.

3.1.1 Overview

The backend is implemented as a **FastAPI** application (`agent/main.py`) that exposes a RESTful API with Server-Sent Events (SSE) for real-time response streaming. The frontend is a **Streamlit** application that communicates with the backend over HTTP, consuming streamed responses. Authentication is handled via **Supabase Auth** using JWT tokens, which are validated on each request through a `HTTPBearer` dependency in FastAPI.

3.1.2 Agent Layer

The core agent is implemented as a LangGraph `StateGraph` (see Figure 3.1). The graph has three nodes:

- `init` — Injects the system prompt as the first `SystemMessage` on a new conversation; is a no-op on resumption.
- `call_llm` — Assembles the full input payload (system prompt, factsheet context, conversation history, and any per-request attachment content), sanitizes it for provider compatibility, then streams the LLM response.
- `call_tool` — Executes all tool calls emitted by the LLM in the previous turn, with per-call caching to avoid redundant invocations within the same session.

Control flows as: `START` → `init` → `call_llm`. A conditional edge then routes to `call_tool` if the LLM response contains tool calls (looping back to `call_llm` afterwards), or to `END` if not. To manage long conversations within the model’s context window, a rolling summarization strategy is applied every `sum_rate` messages (configurable, default 20) using a dedicated `Summarizer` module.

Conversation state is persisted using LangGraph’s `AsyncPostgresSaver` checkpointer, backed by **Supabase PostgreSQL**, enabling session resumption and multi-turn dialogue across requests.

3.1.3 Tool Layer

The agent has access to a set of callable tools defined in `tools.py`. The full tool set (`TOOLS`) used in the primary agent configuration includes:

- `tavily_search` — Web search via the Tavily API.

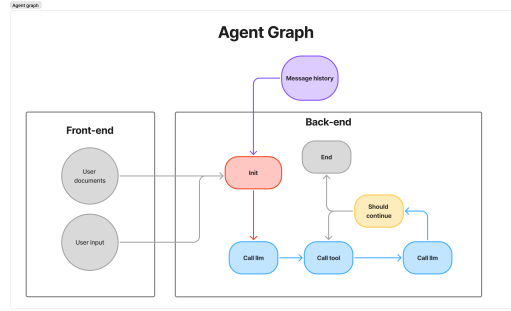


Figure 3.1: Agent graph: `init` → `call_llm` with a conditional loop through `call_tool`.

- **read_attachments** — Fetches and returns the full parsed content of one or more documents by file ID, reading from a database body cache when available and falling back to GCS otherwise.
- **list_attachments** — Lists a project’s attachments and emails (with metadata such as title, date, and category), optionally filtered by date range and significance. Returns file IDs that the model can pass to **read_attachments**.
- **show_elements** — Returns structured factsheet elements (events, parties, claims, damages, deadlines) for a project, optionally filtered by date range and significance level.
- **query_project_attachments** — Performs semantic retrieval (RAG) against project-specific document chunks indexed in BigQuery Vector Store.
- **query_laws** — Semantic retrieval against a corpus of Norwegian legislation, with optional filtering by law title.
- **read_specific_law** — Retrieves exact paragraph text from legislation by title and paragraph number via a parameterised BigQuery query.

An ablated tool configuration, **BASELINE_RAG_TOOLS**, is defined for evaluation purposes. It includes web search, law retrieval (**query_laws**, **read_specific_law**), and project document RAG (**query_project_attachments**), but excludes the structured factsheet tools and direct file-reading access.

3.1.4 Data and Storage Layer

Document persistence is handled through two complementary mechanisms. Raw files (PDF, DOCX, PPTX, EML) are stored in **Supabase Storage**, where they can be retrieved on demand by the **read_attachment** tool. Parsed and chunked document content is embedded using **Google’s gemini-embedding-001 model** and stored in **BigQuery Vector Store** for semantic retrieval. Structured project data — including the **FactSheet** and session metadata — is stored in Supabase PostgreSQL via the **SupabaseManager** CRUD interface.

3.1.5 Init and Update Pipeline

Project initialization (**POST /init-project**) and updates (**POST /update-project**) are handled by the **ProjectPipeline** class, which compiles a LangGraph **StateGraph** for each operation (see Figure 3.2). The graph exploits LangGraph’s fan-out semantics to run independent work in parallel:

- **collapse_emails** and **initialize_input** start in parallel from **START**. **collapse_emails** merges EML thread duplicates; **initialize_input** runs an LLM call to extract an initial structured summary from the user’s case description.
- **extract_emails** follows **collapse_emails** and decodes email attachments from the collapsed threads.
- **parsing** and **storage** both fan out from **extract_emails**. **parsing** processes all documents (PDF/OCR, DOCX, PPTX, EML) into plain text; **storage** uploads raw files to GCS in parallel.

- **embedding** follows **parsing** and indexes chunked document content into BigQuery Vector Store.
- **analyze** is a fan-in node that waits for both **parsing** and **initialize_input** to complete. It then dispatches all documents and emails to the LLM in size-bounded batches (see Section 3.2) to extract structured entities: attachments, emails, events, claims, damages, and deadlines.
- **update_metadata** follows **analyze** and runs an LLM call to update the case title and background based on all extracted data.
- **qc_analysis** also fans out from **analyze** and runs a quality-control pass over the extracted entities.
- **save** follows **update_metadata** and persists all extracted entities to Supabase PostgreSQL.

The update pipeline is identical in structure but adds a **load_project_data** node at the start that loads the existing factsheet before analysis, enabling incremental updates. Status events are streamed back to the frontend via SSE at each node transition.

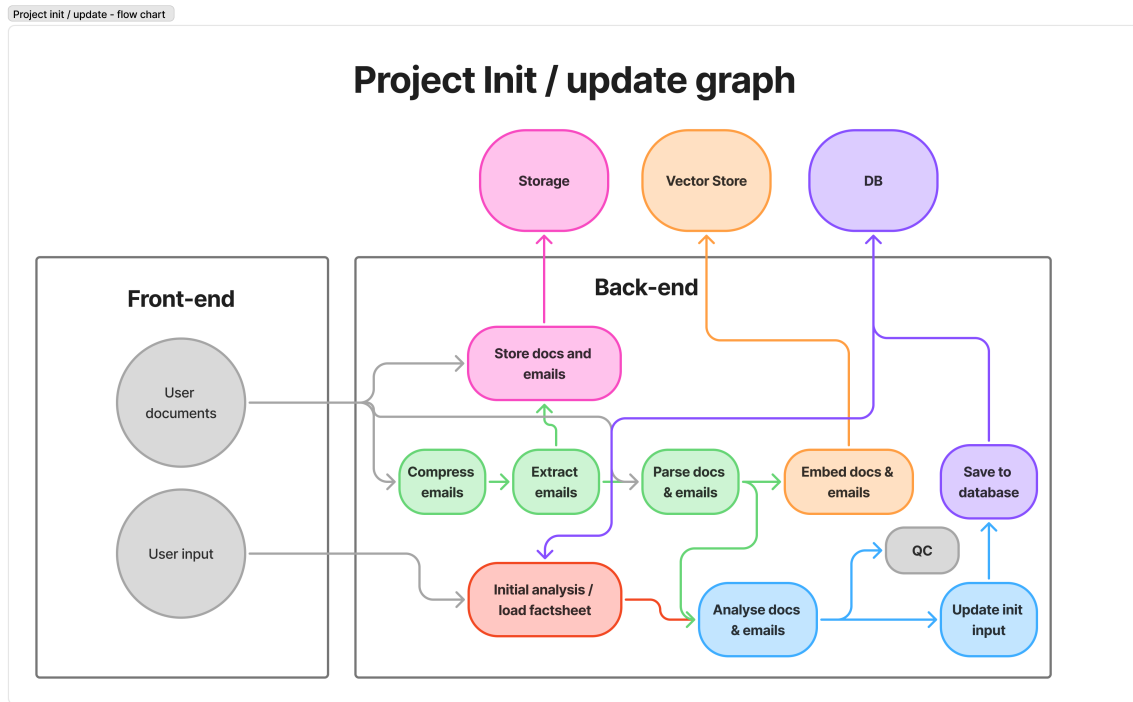


Figure 3.2: Init and update pipeline graph: parallel fan-out for parsing, storage, and initialisation, converging at the **analyze** fan-in node.

3.1.6 Clean Pipeline

The **ProjectClean** class provides two separate cleanup graphs that can be triggered on demand (see Figure 3.3):

- **Clean Elements** (`compile_clean_elements`): **load_project** → **clean** → **save_results**. The **clean** node calls **deduplicate_elements** on the **ContextManager**, which sends all element types (events, parties, claims, damages, deadlines) to the LLM in configurable chunks via **asyncio.gather**. The LLM returns the IDs to keep; all others are discarded. Results are saved back to the database concurrently.
- **Clean Metadata** (`compile_clean_metadata`): **load_project** → **clean** → **save_results**. The **clean** node rewrites the case title and background using the current factsheet content as context. Title and background are saved back concurrently via **asyncio.gather**.

Both graphs share the same linear `StateGraph` structure and reuse the `_load_project_node` and `save` nodes, differing only in the `clean` step.

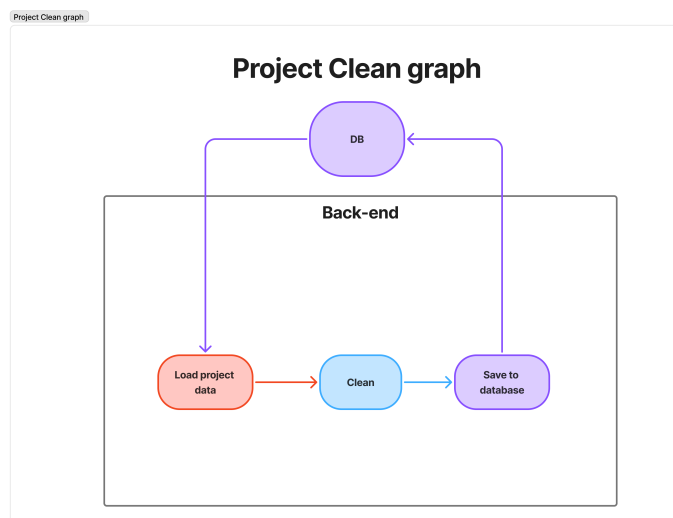


Figure 3.3: Clean pipeline graph: `load_project` → `clean` → `save_results`, used for both element deduplication and metadata rewriting.

3.1.7 LLM Configuration

The system supports multiple LLM providers through LangChain’s `init_chat_model` interface. **Google Gemini** serves as the primary provider, with **OpenAI GPT** available as a secondary option. The active model is configurable per session request, allowing for flexible experimentation and evaluation across providers.

3.1.8 Tech Stack Summary

Layer	Technology
Language	Python 3.13
Agent framework	LangGraph (StateGraph)
LLM orchestration	LangChain (<code>init_chat_model</code>)
LLM providers	Google Gemini (primary), OpenAI GPT (secondary)
Backend API	FastAPI + Uvicorn
Streaming	Server-Sent Events (SSE)
Frontend	Streamlit
Authentication	Supabase Auth (JWT)
Database	Supabase (PostgreSQL)
Vector store	BigQuery Vector Store (persistent)
File storage	Supabase Storage
Conversation checkpointer	LangGraph <code>AsyncPostgresSaver</code>
Document parsing	PyPDF2, ocrmypdf, python-docx, python-pptx
Embeddings	Google <code>gemini-embedding-001</code>
Web search	Tavily

Table 3.1: System tech stack.

3.2 Design Choices

The system architecture was guided by several pragmatic design choices, prioritizing development speed, flexibility, and cost-efficiency given the constraints of a master’s thesis. By separating the descriptive architecture from the rationale, the core motivations for the selected technologies are outlined below.

Agent Layer and Orchestration

The core agent is implemented using **LangGraph** [Lan24]. This framework was chosen primarily for its fast build time, which is essential given the limited time available for this thesis. It provides a clean **StateGraph** abstraction that is highly expandable for future extensions, such as adding dedicated reasoning nodes or document re-ranking stages. Furthermore, LangGraph is model-agnostic, providing built-in support for multiple LLM providers. This flexibility is crucial, allowing the system to easily switch between models or add new ones as desired for evaluation and experimentation.

Database and Storage Architecture

The system’s persistence layer separates structured relational data, unstructured document storage, and semantic vector retrieval to optimize for specific access patterns and constraints.

Relational Storage (PostgreSQL via Supabase): PostgreSQL was selected for its flexibility and highly efficient reads across tables. The structured nature of the factsheet state requires strict enforcement of relationships through primary and foreign keys to maintain data integrity and orderly matching on IDs [Sup20]. An initial attempt was made using a NoSQL database (Firestore); while easy and fast to set up, the lack of an enforced schema led to a loss of overview, with state inconsistencies and errors accumulating over time. Additionally, performing complex reads and combining data across different entities proved expensive and inefficient in NoSQL.

Supabase was chosen as the PostgreSQL provider because it is affordable, open-source, and seamlessly integrates other necessary products like Authentication. Having these features included out-of-the-box makes the development process straightforward and fast.

Database Schema

The database schema is designed to support the agent’s state management and document metadata, while being flexible enough to accommodate future extensions. There are three main groupings of tables:

Project related tables
Projects
Parties
Events
Deadlines
Attachments
Emails
Claims (legal specific)
Damages (legal specific)

Session related tables
Sessions
Session Events
Attachments

Tables for storing document metadata and relationships.

Agent checkpoints
Checkpoints
Writes
Blobs

Tables for the agent’s memory and conversation history.

These tables are combined to form the factsheet state, which is the agent’s internal representation of the case. The factsheet is updated and referenced throughout the agent’s operation, and it is crucial that it is stored in a structured and consistent manner.

File Storage (Supabase Storage): Raw documents (PDF, DOCX, EML, etc.) are stored in Supabase Storage. There were no specific architectural requirements for this other than cost-efficiency. Since Supabase was already selected for the PostgreSQL database, using its built-in, affordable object storage was a natural and convenient choice [Sup20].

Vector Store and Embeddings: Semantic retrieval relies on **Google’s embedding model** (`gemini-embedding-001`). This model was specifically chosen for its strong multilingual capabilities; smaller and cheaper open-source models often struggle with RAG matching when queries or documents are not strictly in English, which is a critical requirement for handling Norwegian legal texts.

For the vector database, a two-tier approach is used. **ChromaDB** is utilized as an in-memory store for fast, session-scoped retrieval. For persistent, project-scoped retrieval, **BigQuery Vector Search** is employed. BigQuery was chosen because it is affordable and operates on an on-demand pricing model. The primary drawback of BigQuery is its high latency, meaning a significant amount of time is spent indexing and saving to the vector store. However, this trade-off is worthwhile: the alternative would require provisioning an always-running vector database server, which introduces high continuous costs and additional operational setup.

FastAPI and Server-Sent Events

The backend is exposed as a standalone REST API using **FastAPI** [Ram18] to decouple it from the frontend. **Server-Sent Events (SSE)** are used for streaming responses because the agent pipeline is long-running and asynchronous. SSE provides a responsive user experience by pushing incremental status updates and partial results without the complexity of WebSockets, which are unnecessary for this unidirectional communication flow.

Concurrency, Reliability, and Tool Efficiency

Several engineering decisions were made to ensure the system operates efficiently under the latency and throughput constraints imposed by external APIs and large document volumes.

Async-first architecture. The entire backend stack is built on Python’s `asyncio` event loop. FastAPI natively supports async request handlers, and all LangGraph pipeline nodes are declared as `async` coroutines. This allows the server to handle multiple concurrent sessions without blocking on I/O-bound operations such as LLM calls, database queries, or storage reads.

Semaphore-based concurrency control. To prevent overloading external services, four independent `asyncio.Semaphore` instances are used — one per resource class: LLM, database, storage, and vector store. Each semaphore has a configurable maximum concurrency value drawn from `config.toml` (e.g. `[async.llm] max_concurrent_requests = 20`). Pipeline nodes acquire the relevant semaphore before dispatching work, ensuring that bursts of parallel tasks are automatically throttled without busy-waiting or sleep loops.

Batch-wise and chunk-wise parallel processing. During project initialization, documents are grouped into batches before being dispatched for LLM analysis. Batching is driven by two configurable thresholds: a maximum byte-size per batch (`project.threshold`) and a maximum file count per batch (`project.max_attachments / project.max_emails`). Each batch is wrapped in an async task that acquires the LLM semaphore, and all tasks are dispatched concurrently via `asyncio.gather()`. The same chunk-wise strategy is applied during factsheet cleaning and deduplication: element lists (events, claims, damages, etc.) are split into fixed-size chunks (`project.clean.chunk_size`), each chunk is cleaned by a separate concurrent LLM call, and results are merged after all tasks complete. This approach keeps individual prompts within token limits while maximising throughput.

Rate limiting and 429 handling. To comply with provider-imposed rate limits and avoid HTTP 429 (Too Many Requests) errors, each LLM instance is wrapped with LangChain’s `InMemoryRateLimiter`. The limiter implements a token-bucket algorithm: tokens are refilled at a configurable rate (`async.llm.requests_per_sec`).

and a burst capacity (`async.llm.max_burst_size`) allows short spikes. A single stateful limiter instance is shared across all coroutines that use the same model, so the rate limit is enforced globally rather than per-request.

Retry mechanism with exponential backoff. Transient API errors (e.g. network timeouts or momentary 5xx responses) are handled by a dedicated `apply_retry()` helper. It wraps any LangChain chain with `with_retry()`, configuring a stop condition (`retry_attempts` from config) and exponential backoff with random jitter (`wait_exponential_jitter=True`). The retry decorator is applied *after* structured-output binding so that validation failures are also covered. If all retry attempts are exhausted, the error is propagated and logged.

TOML-based configuration management. All tunable parameters — concurrency limits, rate limiter settings, retry counts, chunk sizes, model endpoints, and storage bucket names — are centralised in a single `config.toml` file. On startup, the file is parsed into a fully typed `AppConfig` object built with `Pydantic` and `pydantic-settings`. Nested configuration sections map directly to typed dataclasses (e.g. `AsyncConfig`, `ProjectConfig`, `VectorstoreConfig`), providing IDE auto-complete, validation at load time, and a single source of truth for the entire application. A `get_app_config()` singleton cached with `@lru_cache` ensures the config is parsed only once per process.

Tool interface design — string IDs over full objects. Agent tools are designed to accept minimal, flat inputs: all tools receive only string identifiers (e.g. `project_id: str, ids: list[str]`) rather than full domain objects. This keeps the JSON schema presented to the LLM small, reduces the token footprint of tool calls, and makes it straightforward for the model to construct valid invocations. Internal data fetching (loading full attachment records, querying the database) is performed inside the tool implementation, hidden from the model entirely.

Context compression via filtering and display shortening. A key design concern for the custom agent is managing the size of the structured context injected into each LLM call. As a case accumulates material, a naive approach of injecting the full factsheet verbatim would cause token consumption to grow unboundedly. Two complementary compression mechanisms are applied to mitigate this.

First, the `show_elements` and `list_attachments` tools accept optional `significance` and `date_range` parameters. These filters are implemented as PostgreSQL RPC functions on the Supabase backend and are applied server-side before any data is returned to the agent, ensuring that only elements above a configurable significance threshold and within the relevant time window are included in the context.

Second, the string representation of each structured element is deliberately abbreviated. Rather than serialising the full object with all fields, a compact display format omits less informative fields and truncates long free-text descriptions. Together, these two mechanisms reduce the effective model input by approximately 70–80% compared to an unfiltered, full-fidelity serialisation of the same factsheet. This compression is a significant contributor to the finding that the custom agent’s average token consumption remains statistically indistinguishable from that of the simpler baseline agent despite carrying substantially richer case context.

3.3 Evaluation Design

Describe the evaluation design, including the datasets used, metrics for evaluation, and the experimental setup.

3.3.1 Datasets

The evaluation relies on two datasets developed specifically for this thesis in collaboration with legal experts from the law firm BAHR AS. Each dataset represents a real legal case processed through the Norwegian court system.

THRD-2021 is based on a publicly available district court case. The authentic court record — comprising the final submissions from both parties and the court’s verdict — is used directly as part

of the document set. The remaining case materials (correspondence, internal notes, technical reports, agreements, and supporting evidence) were not publicly available and were therefore synthetically reconstructed to match the factual matrix and legal reasoning documented in those authentic records. This hybrid approach provides a realistic document topology while respecting confidentiality.

TOSL-2024 is a fully authentic case for which explicit permission was obtained to use the original documents. It is substantially larger and more complex than THRD-2021: it involves more parties, a longer timeline, more contested legal issues, and a higher total document volume. Because TOSL-2024 is a real case with real documents, it more closely represents the conditions under which the agent would operate in production, and its results should therefore be given greater interpretive weight than those from THRD-2021.

Both datasets are organized to simulate the chronological lifecycle of a legal proceeding. Each case is structured into several *sessions*, each containing a sequence of *queries*. This structure is designed to evaluate two critical capabilities of the agent:

The datasets are organized to simulate the chronological lifecycle of a legal proceeding. Each case is structured into several *sessions*, each containing a sequence of *queries*. This structure is designed to evaluate two critical capabilities of the agent:

1. **Contextual Awareness:** The ability to maintain a coherent understanding of the case as it evolves through different stages of litigation.
2. **Analytical Accuracy:** Each query is paired with a "gold standard" expected output, defined by legal experts. These references serve as the ground truth for measuring correctness, relevancy, and completeness.

By following a chronological order, the sessions require the agent to handle increasing amounts of information and maintain consistency over time, reflecting the professional requirements of a legal practitioner managing an ongoing case.

3.3.2 Dataset Limitations and Caveats

Several factors limit the generalisability of the evaluation and should be considered when interpreting the results.

Absence of a plain LLM baseline. A natural comparison point would be an ordinary LLM queried without any specialised architecture — no document processing, no vector store, no RAG, simply the raw case files passed as context. This approach was explicitly attempted. In practice, the aggregated size of a realistic legal case file — spanning contracts, correspondence, technical reports, and legal submissions — immediately exhausted the rate limits of every model tested, well before a meaningful response could be produced. The sheer volume of document content triggers the providers' token-per-minute limits even for a single query, making repeated evaluation across sessions entirely infeasible. Feeding entire case files into a plain LLM context is therefore not a viable strategy for any of the models evaluated, and the minimum meaningful comparison is a RAG-enabled architecture. This is also why the evaluation compares the custom agent against a Baseline + RAG configuration rather than an unaugmented LLM.

Dataset size and the scalability trade-off. The evaluation uses a constrained number of cases and queries, as running a large-scale benchmark across multiple model-agent pairs with repeated trials becomes prohibitively expensive and time-consuming. This is, however, in tension with a fundamental property of the system: the agent is designed to improve with larger and richer document sets, as more material increases the factsheet's coverage and the quality of RAG retrieval. Evaluating on small datasets therefore likely *underestimates* the performance advantage of the custom agent in production-scale deployments.

Heterogeneity between cases. The two evaluation datasets differ substantially in origin, complexity, and document volume. THRD-2021 uses a hybrid document set (authentic court record supplemented by synthetically reconstructed case materials), while TOSL-2024 consists entirely of real documents. TOSL-2024 is considerably more complex: it covers more parties, a longer contested timeline, and a larger total document volume. This asymmetry means that performance differences across

datasets may partly reflect document complexity rather than agent capability alone. However, since the agent is specifically designed for complex, large-scale cases, TOSL-2024 is the more ecologically valid test of its capabilities, and the dataset-level breakdowns in the results should be read with this in mind.

Limitations of expected-answer evaluation. Model responses are graded against expected answers produced in collaboration with legal experts at BAHR AS. This approach carries two inherent risks: the expected answers themselves may contain inaccuracies, and — more fundamentally — many legal queries are open-ended, such that a correct and complete model response may differ substantially from the reference without being wrong. Three mitigations were applied to reduce this risk:

1. **Fact-based scope:** The evaluation is restricted to factual, document-grounded questions (e.g. "Who are the parties?", "What damages are claimed?"), deliberately excluding normative or predictive legal questions where correct answers are inherently ambiguous.
2. **Validatable content:** Questions are formulated so that their answers can be directly verified against the case documents, reducing the space for legitimate variation.
3. **Minimal-requirement framing:** Expected answers are written as *minimum* acceptable responses rather than exhaustive ideal answers, so a model response that covers the required facts is scored as correct even if it includes additional relevant information.

3.3.3 Evaluation Environment

Evaluation Framework:

The evaluation will be conducted using **DeepEval** [Con23], a framework for evaluating LLM outputs against reference answers. DeepEval allows for automatic scoring based.

Agent Configurations:

Two agent configurations are evaluated:

- **Custom Agent:** The full agent configuration with access to all tools, including structured factsheet access (`show_elements`, `list_attachments`), direct file reading (`read_attachments`), and RAG capabilities.
- **Baseline + RAG Agent:** An ablated version with access to web search, law retrieval, and project document RAG, but without direct file access or structured factsheet tools.

A plain LLM baseline without document retrieval was considered but excluded from the evaluation for the reasons described in Section 3.3.2: the volume of case material immediately saturates provider rate limits, making repeated evaluation impossible.

Model Configuration:

Since the scope of the thesis is not to evaluate the models themselves, but rather the design and architecture of the agent, a selection of four models is used for comparison. These include both closed-source and open-source variants to ensure the agent’s performance is consistent across different provider architectures.

Table 3.2: Overview of Large Language Models used in evaluation.

Model Name	Type	Total Params	Active Params	Provider
Claude-sonnet-4-6	Closed	Unknown	Unknown	Anthropic
Gemini 2.5 Flash	Closed	Unknown	Unknown	Google
Qwen 3.5 [YLY+25]	Open (MoE)	397B	17B	Alibaba
Qwen 3 Next Instruct [YLY+25]	Open (MoE)	80B	3B	Alibaba

The evaluation will be conducted across all datasets using consistent model-agent pairings. To account for the inherent stochastic nature of LLM outputs, each evaluation set (model + agent configuration) will be executed 5 times to ensure statistical significance in the results.

3.3.4 Metrics

The evaluation will focus on several key metrics, measuring both the quality of the agent’s responses and its efficiency in terms of resource usage. Metrics that are to be evaluated are:

- **Correctness, Relevancy, and Completeness:** These semantic metrics are evaluated using the DeepEval framework. The agent’s output is compared against a ”gold standard” reference created in collaboration with legal experts. Each metric is scored on a scale from 0 to 1.
- **Self-consistency:** The degree to which the agent’s output remains consistent across multiple queries and over time as the case evolves.

$$\text{Self-Consistency} = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n \cos(\mathbf{e}_i, \mathbf{e}_j) \quad (3.1)$$

where n is the number of generated outputs, $\mathbf{e}_i$ and $\mathbf{e}_j$ are the sentence embeddings of the i -th and j -th output, and the cosine similarity is defined as:

$$\cos(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|} \quad (3.2)$$

- **BERTScore:** A metric that evaluates the similarity between the agent’s output and a reference text using BERT embeddings [ZKW⁺20].
- **Passrate:** The percentage of queries for which the agent’s output meets a predefined quality threshold.
- **Efficiency Metrics:** **Token Usage** (total tokens consumed) and **Inference Time** (latency) are recorded to assess the practical viability of the system.

Correctness, relevancy, Completeness are all evaluated against the expected out (constructed in collaboration with legal experts) using DeepEval. The framework rates each response on a scale from 0 to 1, with a certain threshold for what is considered a ”pass” for the passrate metric.

BERTScore is calculated based on the expected output. Self-consistency is evaluated by comparing the agent’s responses across multiple queries and sessions to assess whether it maintains a coherent understanding of the case over time. Token usage and inference time are recorded during the agent’s operation.

3.3.5 One-sided Hypothesis Tests

Inspection of the metric distributions reveals pronounced right-skew with heavy concentration near 1.0 (see Figure 3.4), violating the normality assumption required by parametric tests such as the t -test. All hypothesis tests are therefore conducted using the one-sided **Mann-Whitney U test**, a non-parametric rank-based test that assesses whether one distribution tends to produce stochastically larger values than another, without assuming normality.

For quality/performance metrics (correctness, relevancy, completeness, passrate, self-consistency, BERTScore):

$$H_0 : \eta_{\text{custom}} \leq \eta_{\text{baseline+RAG}} \quad H_1 : \eta_{\text{custom}} > \eta_{\text{baseline+RAG}}$$

For cost/efficiency metrics (token usage, inference time):

$$H_0 : \eta_{\text{custom}} \geq \eta_{\text{baseline+RAG}} \quad H_1 : \eta_{\text{custom}} < \eta_{\text{baseline+RAG}}$$

Notation:

- η_{custom} = population median for the custom agent
- $\eta_{\text{baseline+RAG}}$ = population median for the baseline with RAG agent

All tests are one-tailed / one-sided.

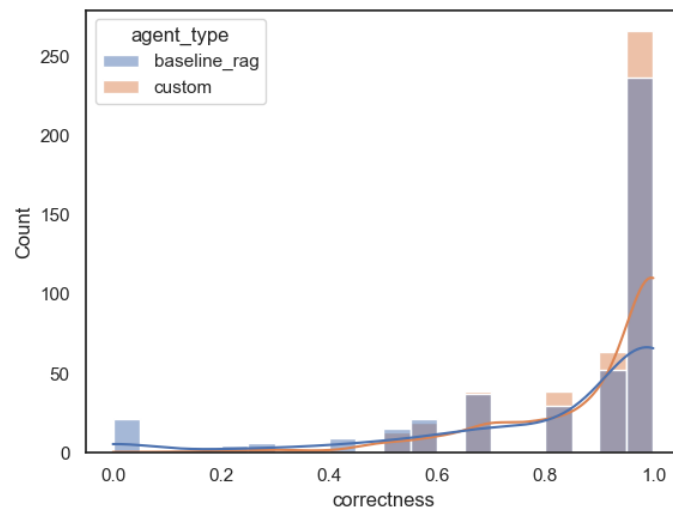


Figure 3.4: Distribution of correctness scores across all queries and sessions. The pronounced right-skew and heavy concentration near 1.0 violate the normality assumption, motivating the use of non-parametric Mann-Whitney U tests.

Chapter 4

Results

4.1 Expected Performance Patterns

Before presenting the empirical results, this section outlines the a priori expectations that motivate the experimental design. These expectations are grounded in the architectural differences between the two agent configurations and in the nature of legal case progression.

Quality metrics over session progression. The central hypothesis is that the performance trajectories of the two architectures diverge as a case progresses. In the earliest sessions, the document set is small and the total context volume is modest. Under these conditions the less complex agent—which does a simple retrieval based on a query—performs well due to the manageable context size, as the complexity of the task is low. The difference between the two architectures is expected to be minimal.

As subsequent sessions introduce new documents, new parties, additional claims, and a growing timeline, the situation reverses. The simpler RAG approach becomes increasingly more difficult as the returned content not necessarily contains sufficient context to answer the question where the answer lies spread across multiple documents. The agent is forced to ingest an ever-larger payload, increasing the risk of retrieval confusion, cross-document interference, and eventual context overflow or to answer on a poor ground, reducing the overall performance. The custom agent, by contrast, maintains a continuously updated factsheet and performs targeted retrieval against a variety of tools and structured information, as well as the same vectorstore as the baseline, allowing it to leverage accumulated case knowledge rather than being overwhelmed by it. The expected result is a positive trajectory for the custom agent across sessions and a flat or degrading trajectory for the baseline.

Token usage over session progression. Token usage is expected to grow for both architectures as the case accumulates material, but for structurally different reasons. For the baseline agent, the number of input tokens often increases with the case size, as the retrieval search requires context from different sources and accumulates over time when a lot of queries are done and each returns multiple chunks of whom many are less relevant. For the custom agent, input token growth reflects increasing *richness* rather than raw volume: the factsheet grows denser, the returned context are compressed and the model are giving a variety of options to filter only the necessary. Output tokens are likewise expected to grow over time for the custom agent, as richer context enables more thorough and nuanced answers. This growth is therefore a positive signal—evidence that the agent is leveraging accumulated case knowledge—rather than an efficiency cost.

Inference time. Response latency is expected to increase with case progression for both architectures.

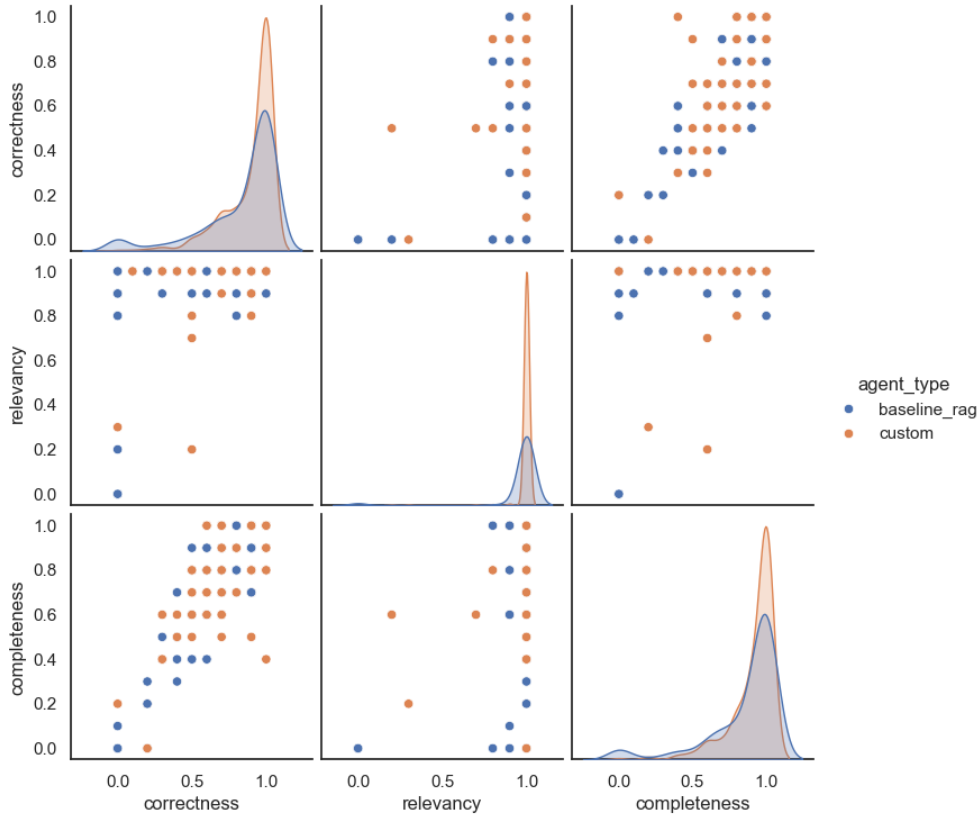


Figure 4.1: Pairplot of DeepEval quality metrics (correctness, relevancy, completeness). Diagonal panels show per-metric density estimates. The custom agent (orange) exhibits higher correctness and completeness, while relevancy is similarly high for both configurations.

4.2 Quality Metrics

DeepEval Quality Metrics (Aggregate)

Figure 4.1 shows the joint distribution of correctness, relevancy, and completeness across all queries and sessions, aggregated for each agent type.

The custom agent’s correctness distribution (orange) is clearly left-skewed, with density concentrated near 1.0, whereas the baseline distribution (blue) is considerably more spread. This confirms that the correctness advantage observed in the longitudinal plot is not driven by a small number of high-scoring queries but reflects a broad shift in the distribution.

Relevancy scores are highly concentrated near 1.0 for both configurations, indicating that both agents are generally capable of producing on-topic responses. This suggests that the divergence in correctness is attributable to factual accuracy rather than off-topic generation.

Completeness shows a similar pattern to correctness: the custom agent’s distribution skews higher, while the baseline exhibits more low-completeness responses. This is consistent with the expectation that the factsheet provides richer context for constructing comprehensive answers, especially as the case grows in scope.

Overall, the pairplot reveals the increase in consistency and reliability of the custom agent’s responses across all three quality dimensions, while the baseline’s performance is more variable and often falls short.

Figure 4.2 shows the trajectories of the three DeepEval metrics across sessions. Both agents broadly track each other through the session progression, but the custom agent remains consistently above the baseline throughout. Crucially, the variance of the custom agent’s trajectory is substantially lower: where the baseline exhibits wider swings between sessions, the custom agent maintains a more stable

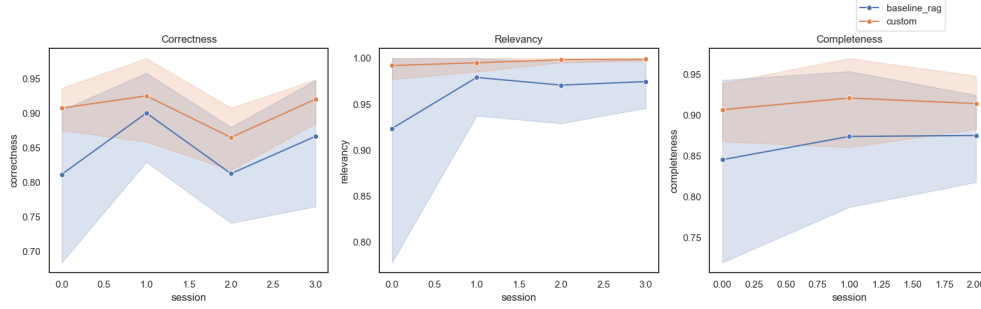


Figure 4.2: Metrics across sessions

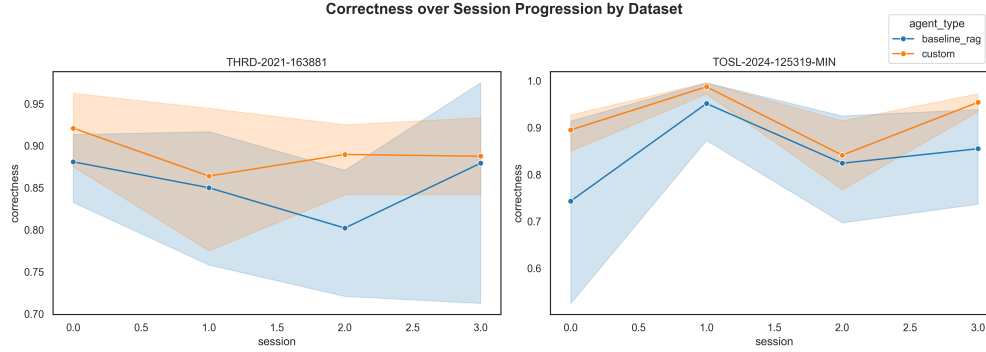


Figure 4.3: Correctness over Session Progression by Dataset

performance profile. This session-level consistency reinforces the pairplot finding that the quality advantage is not driven by occasional strong queries but reflects a broad, reliable improvement.

Figure 4.3 breaks down the correctness trajectory by dataset. The two datasets have distinct characteristics: TOSL-2024 is a fully authentic case with a larger document volume and greater factual complexity, while THRD-2021 contains a mix of authentic court documents and reconstructed case materials (see Section 3.3.2). The dataset breakdown reveals that the performance gap between the two agents is more pronounced on TOSL-2024, where the custom agent’s advantage grows more noticeably as sessions accumulate. On THRD-2021, the custom agent also leads, but the gap is smaller and more stable. This pattern is consistent with the architectural hypothesis: the factsheet and structured retrieval tools confer the greatest benefit precisely when the case is large and complex, where naive RAG retrieval is most prone to missing relevant facts dispersed across many documents.

Figure 4.4 shows the distribution of DeepEval metrics broken down by the underlying LLM. The custom agent outperforms the baseline across all three metrics for every model tested, with one exception: for Gemini 2.5 Pro — the most capable model in the evaluation — the two configurations converge and the difference becomes negligible. For the smaller and less capable models, the margin is wider. This pattern suggests that the structured factsheet context acts as a capability amplifier that has the greatest marginal effect when the base model has limited capacity to compensate for noisy or incomplete retrieval. The implications of this model-dependent effect are discussed further in Chapter 5.

BERTScore

Figure 4.5 shows the joint distribution of BERTScore components (precision, recall, F1, and sentence-BERT similarity). The distributions have considerable overlap between the two agent types, but the custom (orange) shows a higher peak, indicating less variation and more consistent surface-level semantic similarity to the reference answers. The baseline (blue) has a wider spread, with more low-scoring outliers, suggesting that it occasionally produces responses that are less semantically aligned with the reference.

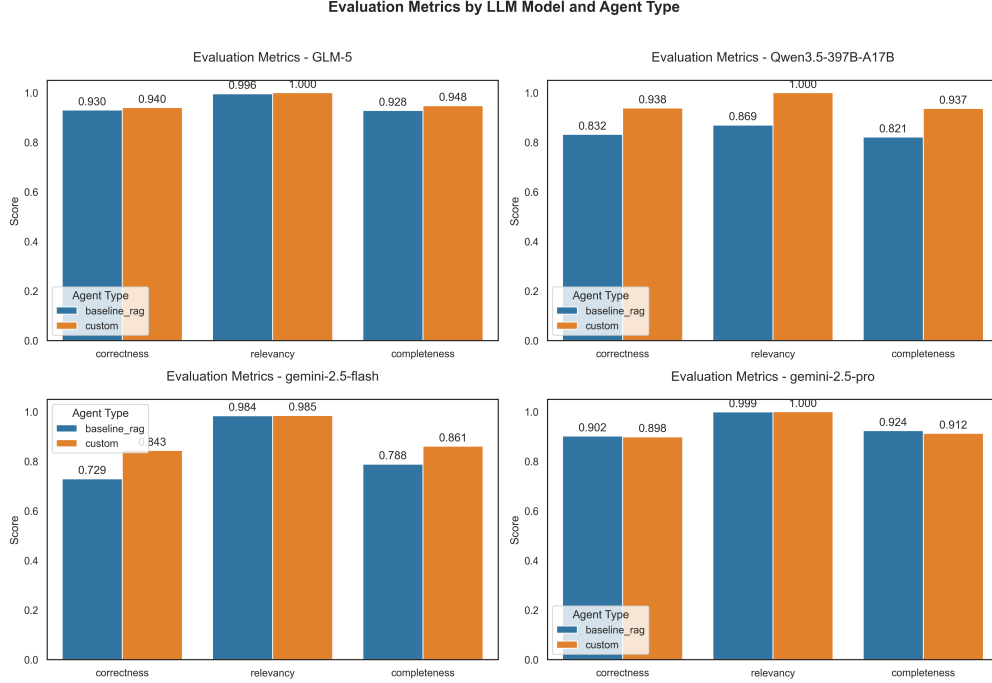


Figure 4.4: Metrics across models

This result is not unexpected. BERTScore measures surface-level semantic similarity to the reference answer and is therefore less sensitive to structured factual completeness than the DeepEval metrics. A response can score well on BERTScore while omitting specific facts that correctness and completeness evaluation would penalise. The near-identical BERTScore distributions therefore complement rather than contradict the DeepEval findings: both agents produce semantically coherent output, but the custom agent more reliably includes the specific factual content required by the gold-standard reference.

Figure 4.6 shows the trajectories of BERTScore F1 and sentence-BERT similarity across sessions. The trajectories for both metrics are relatively flat and closely aligned between the two agent types, with only minor fluctuations. The custom agent (orange) shows a more consistent trajectory with narrower confidence intervals.

4.2.1 Resource Metrics

Figure 4.7 shows the joint distribution of resource consumption metrics: inference duration, input tokens, output tokens, total tokens, and number of LLM calls per query.

The most notable pattern is in input token usage. The custom agent (orange) spans a wider range, with some queries consuming upwards of 200,000 input tokens. This is consistent with the expectation that the factsheet—injected in full on each call—grows denser as the case accumulates material, and that the vector retrieval step returns richer and more numerous chunks. The baseline agent (blue) clusters at lower input token counts, reflecting its simpler retrieval strategy, though it too exhibits high-token outliers corresponding to sessions with large document sets.

Output tokens are similarly distributed, with the custom agent showing a slightly wider tail toward higher values. This is in line with the expectation that a richer context enables more detailed answers.

Inference duration is generally lower and more tightly concentrated for the custom agent, with the baseline exhibiting heavier tails at high latency. The structured retrieval mechanism of the custom agent—bounded by configurable top- k limits—appears to reduce the variance in response time compared to the baseline’s unbounded document ingestion.

LLM call counts are comparable between the two configurations, typically in the range of 1–3 calls per query, indicating that neither architecture requires substantially more agentic steps than the other to produce a response.

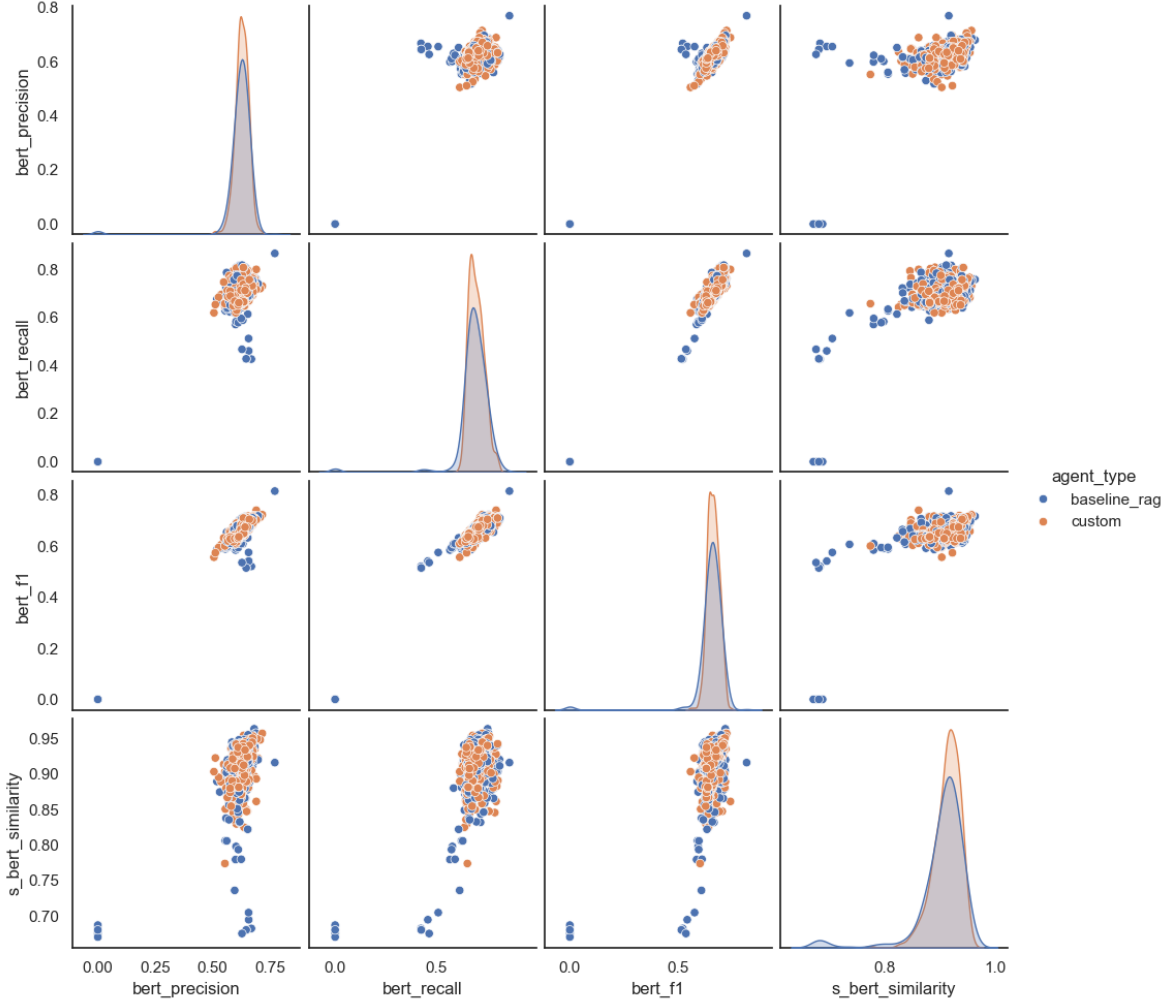


Figure 4.5: Pairplot of BERTScore sub-metrics (precision, recall, F1, and sentence-BERT similarity). Distributions for the two agent configurations largely overlap, indicating comparable surface-level semantic quality while the DeepEval metrics reveal greater divergence in factual completeness.

Figure 4.8 shows the trajectories of resource metrics across sessions. Unlike the quality metrics, no consistent trend is visible for either agent over the session progression: token counts and latencies fluctuate without a systematic increase, and the two configurations track each other closely throughout. This is consistent with the statistical finding of no significant difference in token usage or inference time. The absence of a growth trend is partly explained by the context compression mechanisms applied to the custom agent’s structured context: as the factsheet grows, significance filtering and display shortening keep the payload from expanding proportionally.

4.3 Statistical Modeling

4.3.1 Mann-Whitney U Test

As described in the evaluation design, all hypothesis tests use the one-sided Mann-Whitney U test, which is appropriate given the non-normal distributions of all metrics (Shapiro-Wilk, $p < 10^{-18}$ for every metric tested). The null hypothesis in each case is that the custom agent’s scores are not stochastically greater than those of the baseline agent ($H_0 : \eta_{\text{custom}} \leq \eta_{\text{baseline+RAG}}$).

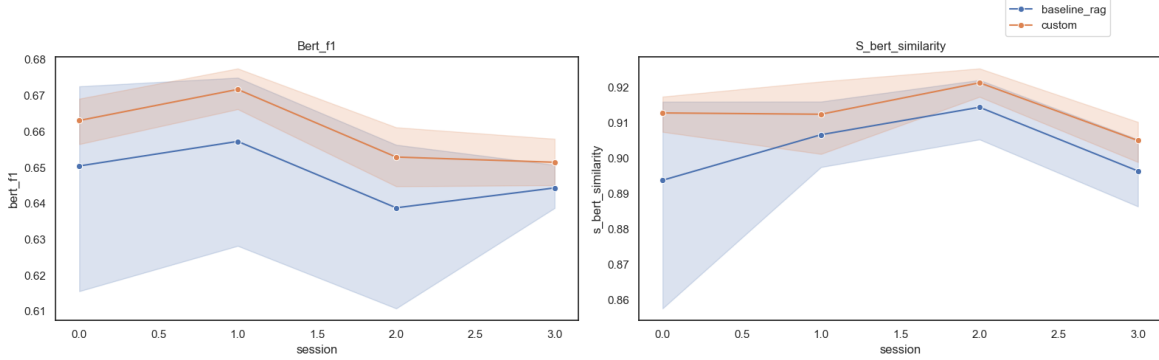


Figure 4.6: Lineplot of BERTScore sub-metrics (precision, recall, F1, and sentence-BERT similarity). Distributions for the two agent configurations largely overlap, indicating comparable surface-level semantic quality while the DeepEval metrics reveal greater divergence in factual completeness.

Quality metrics. The results are summarised in Table 4.1. Correctness and relevancy both show statistically significant advantages for the custom agent at the $\alpha = 0.05$ level. Completeness does not reach significance, although the directional trend is the same.

Table 4.1: Mann-Whitney U test results for quality metrics (one-sided, H_1 : custom > baseline).

Metric	U-statistic	p-value	Result
Correctness	105,587.00	8.49×10^{-3}	Significant ($p < 0.05$)
Relevancy	101,782.50	3.42×10^{-4}	Significant ($p < 0.05$)
Completeness	38,820.00	8.34×10^{-2}	Not significant

The significance of relevancy is noteworthy: given that both agents retrieve from the same underlying vector store, a relevancy advantage suggests that the structured context provided by the factsheet helps the custom agent formulate responses that are more precisely targeted to the question, even when raw retrieval quality is comparable.

The absence of a significant completeness result likely reflects the smaller number of completeness observations in the dataset (completeness evaluation was not collected for all records) and the inherent difficulty of distinguishing partial from complete answers when both agents frequently cover the required facts.

Resource metrics. Testing the directional hypothesis that the custom agent consumes fewer resources than the baseline (H_1 : custom < baseline), neither input token usage ($U = 99,750$, $p = 0.730$) nor inference duration ($U = 93,705$, $p = 0.163$) differs significantly between configurations. Numerically, mean input token counts are almost identical (custom: $72,286 \pm 77,114$; baseline: $71,968 \pm 75,645$) and mean durations are within one second of each other (custom: 35.1 s; baseline: 36.1 s).

This result has an important practical implication: the custom agent achieves its quality improvements without imposing a measurable additional cost in tokens or latency. The context compression mechanisms described in Section 3.2 — filtering by significance and date range via database RPC functions, and display-shortening of structured elements — appear to offset the additional context introduced by the factsheet, keeping total token consumption comparable to the simpler retrieval approach.

4.3.2 Self-consistency

Figure 4.9 shows the distribution of self-consistency scores by agent type. The custom agent (orange) exhibits a higher median self-consistency score compared to the baseline (blue), with a narrower interquartile range, indicating more consistent performance across queries. The baseline agent shows a wider spread of scores, suggesting greater variability in response consistency.

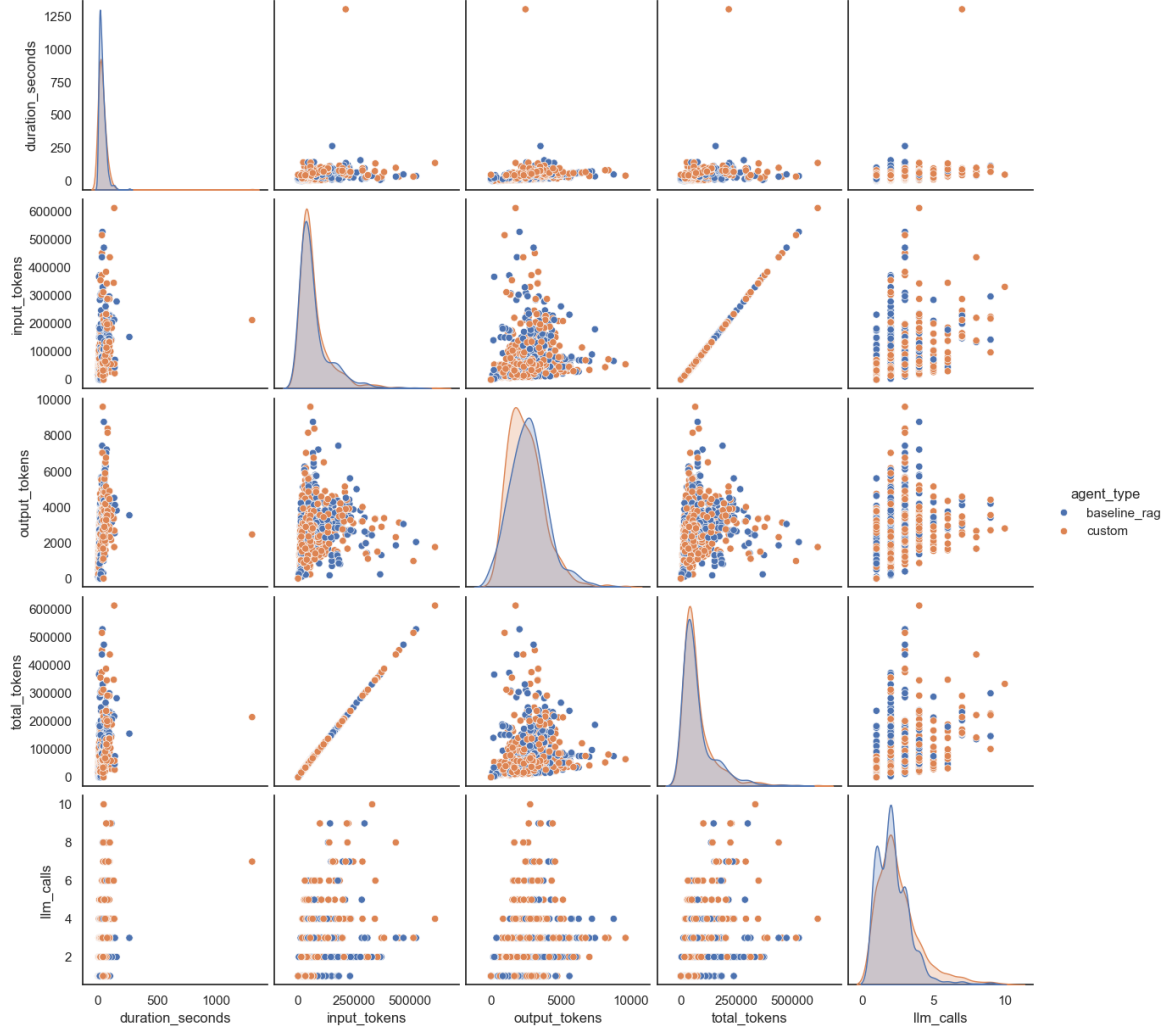


Figure 4.7: Pairplot of resource metrics (duration in seconds, input tokens, output tokens, total tokens, and LLM call count). The custom agent exhibits higher input token usage reflecting factsheet growth, while inference duration is more tightly bounded compared to the baseline.

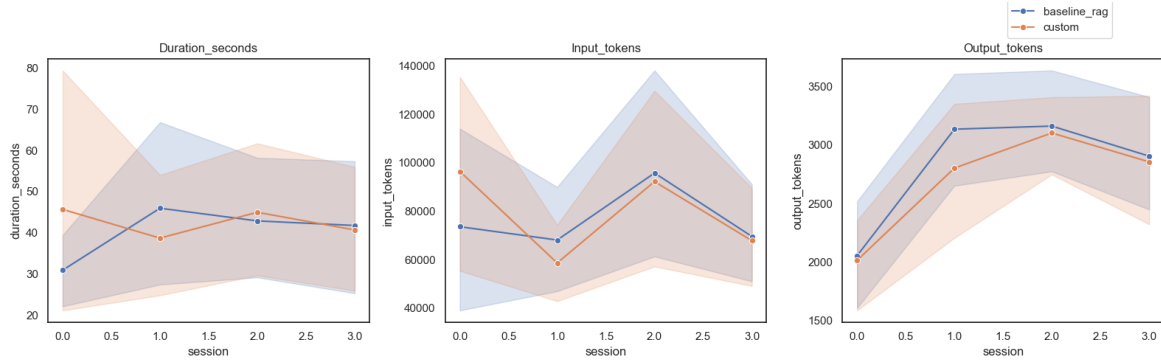


Figure 4.8: Lineplot of resource metrics (duration in seconds, input tokens, output tokens, total tokens, and LLM call count). The custom agent exhibits higher input token usage reflecting factsheet growth, while inference duration is more tightly bounded compared to the baseline.



Figure 4.9: Violin plot of self-consistency scores by agent type.

Chapter 5

Discussion

5.1 Value Beyond Performance Metrics

The evaluation results demonstrate a performance advantage for the custom architecture as case volume grows, but the headline metrics alone understate the practical value of the design. Even in a hypothetical scenario where correctness and completeness scores were identical across configurations, the custom architecture would remain preferable in a production legal setting for reasons that no automatic metric captures.

The factsheet as a product artifact. The baseline agent’s output is ephemeral: it produces a text response and leaves no persistent structured representation of the case behind. The custom agent, by contrast, produces and continuously maintains a structured factsheet as a natural side-effect of normal operation. This factsheet—comprising parties, events, claims, damages, deadlines, and document metadata—is a machine-readable, queryable representation of the case that can be consumed directly by frontend systems without any additional LLM inference. Concrete applications include timeline visualisations, party graphs, claims dashboards, and deadline trackers, all of which can be rendered from the factsheet with no extra cost.

Human oversight and auditability. The structured state enables a qualitatively different relationship between the lawyer and the system. Rather than trusting a black-box text response, a practitioner can inspect the factsheet directly, verify individual entries against source documents, and correct errors at the element level. If the agent produces an incorrect answer, the source of the error can often be traced to a specific factsheet entry—a mis-extracted event date or an omitted party—and corrected there, improving all subsequent responses. This kind of targeted, human-in-the-loop correction is not possible with a stateless RAG agent, where errors are buried in the implicit context and must be addressed by re-uploading or re-prompting.

Versioned case understanding. Because the factsheet is updated incrementally at each session boundary, it constitutes a natural audit trail of how the agent’s understanding of the case evolved over time. Earlier versions can be retained and compared, providing a record of when specific facts, parties, or claims entered the agent’s knowledge base. In legal practice, where the timing of knowledge can itself be legally significant, this version history has potential evidentiary and procedural value.

Downstream system integration. The structured state lowers the integration barrier for adjacent legal tools and workflows. A downstream system can query the factsheet via a standard database interface to, for example, check for conflicts of interest against other open cases, trigger calendar entries for extracted deadlines, or populate court filing templates with party and claim data. These integrations require structured, typed data; they cannot be built on top of free-form agent responses without a costly and fragile natural language parsing layer.

Summary. The performance metrics reported in Chapter 4 capture one dimension of the architecture’s value: how accurately the agent answers factual questions across a progression of sessions. They

do not capture the value of the structured case representation itself—a persistent, inspectable, and reusable artifact that the custom architecture produces as a consequence of its design. In production deployments with real case volumes and real practitioner workflows, this latent value is likely to be at least as significant as the raw performance advantage.

5.2 Uncertainties and Confounding Factors

The evaluation results support the conclusion that the custom agent outperforms the baseline on correctness and relevancy, but several sources of uncertainty affect the magnitude and generalisability of this finding. These are discussed below.

Question difficulty and scope. The evaluation queries were designed to be factual and document-grounded, but the level of specificity varies considerably across sessions. Some questions have unambiguous single-document answers, while others require synthesising information across multiple documents and timelines. Harder queries amplify the architectural advantage of the custom agent — the factsheet and structured retrieval tools are most useful precisely when the answer requires cross-document reasoning — while easier queries may show little to no difference between architectures. The observed aggregate performance gap therefore depends on the composition of the query set: a test set dominated by simple lookups would understate the custom agent’s advantage, while one focused on complex synthesis would overstate it. The current evaluation mix likely sits somewhere in between.

Dataset size and complexity. Only two datasets are used in the evaluation, and their characteristics differ substantially. The results from TOSL-2024, the larger and more complex real case, show a more pronounced advantage for the custom agent than those from THRD-2021. This is consistent with the design hypothesis: the factsheet-based architecture is most beneficial when the document set is large and the case is complex, because those are the conditions under which naive retrieval is most likely to fail. With only two datasets, however, it is difficult to disentangle the effect of document volume from other case-specific characteristics. A larger benchmark spanning more cases and varying document volumes would provide stronger evidence for the scaling behaviour observed here.

Underlying language model capability. Figure 4.4 shows that the performance gap between the custom and baseline agents narrows as the underlying LLM becomes more capable: for the most advanced model tested (Gemini 2.5 Pro), the two configurations are nearly indistinguishable. This suggests two non-exclusive interpretations. First, a sufficiently capable model may compensate for the structural disadvantage of the baseline by inferring missing context or tolerating retrieval noise more robustly. Second, the evaluation queries and datasets may not be sufficiently demanding to expose the limits of a powerful model: the questions can largely be answered from the retrieved chunks even without the accumulated factsheet context. Conversely, the performance gap is most pronounced for the smaller models, for which structured, pre-digested context provides the greatest marginal benefit. This implies that the value of the custom architecture is not fixed but model-dependent: it acts as a capability amplifier that matters most when the underlying model has limited reasoning capacity or context-handling ability.

Stochastic variability. LLM outputs are inherently stochastic. Each evaluation set was run five times to reduce the impact of this variability, and the non-parametric Mann-Whitney U test is robust to the non-normal, right-skewed score distributions observed. Nevertheless, the total number of unique query-session combinations is constrained by the cost and time of running large-scale multi-model evaluations, and the reported p-values should be interpreted with awareness of the limited sample size. The significance of correctness ($p = 8.5 \times 10^{-3}$) and relevancy ($p = 3.4 \times 10^{-4}$) results are robust, but the borderline result for completeness ($p = 8.3 \times 10^{-2}$) should be treated cautiously.

Chapter 6

Conlusion

In progress

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